

Exponentials & Polynomials

ANSWER KEY



Laws of exponents

- 1 Simplify $x^5 \cdot x^3$.

$$x^5 \cdot x^3 = x^{5+3} = x^8.$$

- 2 Simplify $\frac{y^9}{y^4}$.

$$\frac{y^9}{y^4} = y^{9-4} = y^5.$$

- 3 Simplify $(2x^3)^4$.

$$(2x^3)^4 = 2^4 x^{12} = 16x^{12}.$$

Negative and fractional exponents

- 4 Simplify x^{-3} , writing your answer without a negative exponent.

$$x^{-3} = \frac{1}{x^3}.$$

- 5 Simplify $16^{3/4}$.

$$16^{3/4} = (16^{1/4})^3 = 2^3 = 8 \text{ (since } 16^{1/4} = 2, \text{ because } 2^4 = 16).$$

Adding, subtracting, and multiplying polynomials

- 6 Simplify $(3x^2 + 2x - 5) + (x^2 - 4x + 7)$.

$$\text{Combine like terms: } (3x^2 + x^2) + (2x - 4x) + (-5 + 7) = 4x^2 - 2x + 2.$$

- 7 Expand $(x + 3)(x - 5)$.

$$(x + 3)(x - 5) = x^2 - 5x + 3x - 15 = x^2 - 2x - 15.$$

- 8 Expand $(2x - 1)^2$.

$$(2x - 1)^2 = (2x)^2 - 2(2x)(1) + 1^2 = 4x^2 - 4x + 1.$$

Factoring polynomials

- 9 Factor $x^2 - 16$ completely.

This is a difference of squares: $x^2 - 16 = (x - 4)(x + 4)$.

- 10 Factor $2x^2 + 8x$ completely.

Factor out the greatest common factor: $2x^2 + 8x = 2x(x + 4)$.

- 11 Factor $x^3 - 4x$ completely.

Factor out x : $x^3 - 4x = x(x^2 - 4)$, and then factor the remaining difference of squares: $x(x - 2)(x + 2)$.

Polynomial division and the remainder theorem

- 12 Find the remainder when $f(x) = x^3 - 2x^2 + x - 5$ is divided by $(x - 2)$, using the remainder theorem.

By the remainder theorem, the remainder equals $f(2)$: $f(2) = 8 - 8 + 2 - 5 = -3$.

Exponential growth vs. polynomial growth

- 13 Two functions are given: $f(x) = x^3$ and $g(x) = 2^x$. Explain which function eventually grows faster as x increases, and estimate the crossover point using the values $f(10) = 1000$, $g(10) = 1024$, $f(8) = 512$, $g(8) = 256$.

Exponential functions eventually always outgrow polynomial functions as $x \rightarrow \infty$, regardless of the polynomial's degree, because the exponential's rate of increase itself keeps growing while the polynomial's relative rate of increase slows. Comparing the given values: at $x = 8$, $f(8) = 512 > g(8) = 256$ (polynomial ahead), but by $x = 10$, $g(10) = 1024 > f(10) = 1000$ (exponential has overtaken it) — so the crossover happens between $x = 8$ and $x = 10$.

Modeling with exponential functions

- 14 An investment of 5000 dollars grows according to $A(t) = 5000(1.06)^t$, where t is the number of years. Find the value of the investment after 5 years, to the nearest dollar, and state the annual growth rate as a percentage.

$A(5) = 5000(1.06)^5 = 5000(1.3382) \approx 6691$ dollars. The annual growth rate is 6%, since the growth factor is $1.06 = 1 + 0.06$.

Applying the zero product property to polynomial equations

- 15 Solve $x^3 - 5x^2 + 6x = 0$ by factoring completely and applying the zero product property.

Factor out x : $x(x^2 - 5x + 6) = 0$, then factor the quadratic: $x(x - 2)(x - 3) = 0$. By the zero product property, $x = 0$, $x = 2$, or $x = 3$.

Multiplying binomials and trinomials

- 16 Expand $(x + 2)(x^2 - 3x + 4)$.

Distribute each term:

$$x(x^2 - 3x + 4) + 2(x^2 - 3x + 4) = x^3 - 3x^2 + 4x + 2x^2 - 6x + 8 = x^3 - x^2 - 2x + 8.$$

- 17 Expand $(2x - 3)(x + 5)$.

$$(2x - 3)(x + 5) = 2x^2 + 10x - 3x - 15 = 2x^2 + 7x - 15.$$

Comparing exponential models

- 18 Two savings accounts grow according to $A(t) = 1000(1.03)^t$ and $B(t) = 1000(1.05)^t$. Explain which account grows faster, and find the difference in value after 10 years, to the nearest dollar.

Account B grows faster because it has a larger growth factor ($1.05 > 1.03$, meaning a higher annual interest rate). After 10 years: $A(10) = 1000(1.03)^{10} \approx 1344$, $B(10) = 1000(1.05)^{10} \approx 1629$.
Difference: $1629 - 1344 = 285$ dollars.

Simplifying expressions with multiple exponent rules

- 19 Simplify $\frac{(x^3y^2)^2}{x^4y}$.

Expand the numerator: $(x^3y^2)^2 = x^6y^4$. Divide: $\frac{x^6y^4}{x^4y} = x^{6-4}y^{4-1} = x^2y^3$.

- 20 Simplify $\left(\frac{2x^2}{y}\right)^3$.
- $$\left(\frac{2x^2}{y}\right)^3 = \frac{2^3(x^2)^3}{y^3} = \frac{8x^6}{y^3}.$$
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Half-life and decay applications

- 21 A radioactive substance has a half-life of 8 years, starting with 200 grams. Write a model for the remaining amount after t years, and find the amount remaining after 24 years.

Model: $A(t) = 200\left(\frac{1}{2}\right)^{t/8}$. After 24 years: $A(24) = 200\left(\frac{1}{2}\right)^{24/8} = 200\left(\frac{1}{2}\right)^3 = 200\left(\frac{1}{8}\right) = 25$ grams.

Multiple choice — easy 15

22 Simplify x^{12}/x^2 .

a. x^{10} ✓

b. x^{12}

c. x^{11}

d. x^9

$$x^{12}/x^2 = x^{12-2} = x^{10}.$$

23 Simplify $x^6 \cdot x^5$.

a. x^{12}

b. x^{13}

c. x^{10}

d. x^{11} ✓

$$x^6 \cdot x^5 = x^{6+5} = x^{11}.$$

24 Simplify $(x^4)^4$.

a. x^{15}

b. x^{16} ✓

c. x^{17}

d. x^{18}

$$(x^4)^4 = x^{4 \times 4} = x^{16}.$$

25 Simplify $x^9 \cdot x^7$.

a. x^{18}

b. x^{16} ✓

c. x^{17}

d. x^{15}

$$x^9 \cdot x^7 = x^{9+7} = x^{16}.$$

26 Simplify x^{18}/x^9 .

a. x^{10}

b. x^9 ✓

c. x^{11}

d. x^8

$$x^{18}/x^9 = x^{18-9} = x^9.$$

27 Simplify x^{10}/x^2 .

a. x^9

b. x^8 ✓

c. x^7

d. x^{10}

$$x^{10}/x^2 = x^{10-2} = x^8.$$

28 Simplify $x^5 \cdot x^8$.

a. x^{14}

b. x^{15}

c. x^{13} ✓

d. x^{12}

$$x^5 \cdot x^8 = x^{5+8} = x^{13}.$$

29 Simplify $x^{10} \cdot x^8$.

- a. x^{17}
- b. x^{19}
- c. x^{18} ✓
- d. x^{20}

$$x^{10} \cdot x^8 = x^{10+8} = x^{18}.$$

30 Simplify $x^5 \cdot x^6$.

- a. x^{10}
- b. x^{12}
- c. x^{11} ✓
- d. x^{13}

$$x^5 \cdot x^6 = x^{5+6} = x^{11}.$$

31 Simplify x^9/x^2 .

- a. x^6
- b. x^8
- c. x^9
- d. x^7 ✓

$$x^9/x^2 = x^{9-2} = x^7.$$

32 Simplify x^{13}/x^3 .

a. x^{12}

b. x^{10} ✓

c. x^{11}

d. x^9

$$x^{13}/x^3 = x^{13-3} = x^{10}.$$

33 Simplify $x^{10} \cdot x^6$.

a. x^{16} ✓

b. x^{18}

c. x^{15}

d. x^{17}

$$x^{10} \cdot x^6 = x^{10+6} = x^{16}.$$

34 Simplify x^9/x^7 .

a. x^4

b. x^3

c. x^2 ✓

d. x^1

$$x^9/x^7 = x^{9-7} = x^2.$$

35 Simplify $(x^2)^5$.

a. x^{11}

b. x^{10} ✓

c. x^9

d. x^{12}

$$(x^2)^5 = x^{2 \times 5} = x^{10}.$$

36 Simplify $(x^4)^5$.

a. x^{20} ✓

b. x^{22}

c. x^{21}

d. x^{19}

$$(x^4)^5 = x^{4 \times 5} = x^{20}.$$

Multiple choice — medium 15

37 Expand $(4x - 7)(5x + 9)$.

a. $20x^2 + x + 63$

b. $20x^2 + 3x - 63$

c. $20x^2 + x - 63$ ✓

d. $20x^2 - x - 63$

$$(4x - 7)(5x + 9) = 20x^2 + 36x - 35x - 63 = 20x^2 + x - 63.$$

38 Expand $(3x - 6)(2x + 5)$.

a. $6x^2 + 3x - 30$ ✓

b. $6x^2 + 3x + 30$

c. $6x^2 - 3x - 30$

d. $6x^2 + 5x - 30$

$$(3x - 6)(2x + 5) = 6x^2 + 15x - 12x - 30 = 6x^2 + 3x - 30.$$

39 Expand $(x + 9)(5x + 5)$.

- a. $5x^2 - 50x + 45$
- b. $5x^2 + 50x - 45$
- c. $5x^2 + 50x + 45$ ✓
- d. $5x^2 + 52x + 45$

$$(x + 9)(5x + 5) = 5x^2 + 5x + 45x + 45 = 5x^2 + 50x + 45.$$

40 Expand $(3x + 2)(4x + 5)$.

- a. $12x^2 - 23x + 10$
- b. $12x^2 + 23x - 10$
- c. $12x^2 + 23x + 10$ ✓
- d. $12x^2 + 25x + 10$

$$(3x + 2)(4x + 5) = 12x^2 + 15x + 8x + 10 = 12x^2 + 23x + 10.$$

41 Expand $(4x + 9)(4x + 8)$.

- a. $16x^2 + 70x + 72$
- b. $16x^2 - 68x + 72$
- c. $16x^2 + 68x + 72$ ✓
- d. $16x^2 + 68x - 72$

$$(4x + 9)(4x + 8) = 16x^2 + 32x + 36x + 72 = 16x^2 + 68x + 72.$$

42 Expand $(4x + 9)(2x + 4)$.

- a. $8x^2 + 34x + 36$ ✓
- b. $8x^2 + 36x + 36$
- c. $8x^2 + 34x - 36$
- d. $8x^2 - 34x + 36$

$$(4x + 9)(2x + 4) = 8x^2 + 16x + 18x + 36 = 8x^2 + 34x + 36.$$

43 Expand $(5x + 5)(3x - 2)$.

- a. $15x^2 + 5x + 10$
- b. $15x^2 + 7x - 10$
- c. $15x^2 + 5x - 10$ ✓
- d. $15x^2 - 5x - 10$

$$(5x + 5)(3x - 2) = 15x^2 - 10x + 15x - 10 = 15x^2 + 5x - 10.$$

44 Expand $(5x + 6)(3x + 5)$.

- a. $15x^2 + 45x + 30$
- b. $15x^2 + 43x + 30$ ✓
- c. $15x^2 + 43x - 30$
- d. $15x^2 - 43x + 30$

$$(5x + 6)(3x + 5) = 15x^2 + 25x + 18x + 30 = 15x^2 + 43x + 30.$$

45 Expand $(5x - 3)(2x - 1)$.

- a. $10x^2 - 11x - 3$
- b. $10x^2 + 11x + 3$
- c. $10x^2 - 11x + 3$ ✓
- d. $10x^2 - 9x + 3$

$$(5x - 3)(2x - 1) = 10x^2 - 5x - 6x + 3 = 10x^2 - 11x + 3.$$

46 Expand $(5x + 9)(3x + 8)$.

- a. $15x^2 + 69x + 72$
- b. $15x^2 + 67x + 72$ ✓
- c. $15x^2 - 67x + 72$
- d. $15x^2 + 67x - 72$

$$(5x + 9)(3x + 8) = 15x^2 + 40x + 27x + 72 = 15x^2 + 67x + 72.$$

47 Expand $(2x + 8)(4x - 3)$.

- a. $8x^2 + 26x + 24$
- b. $8x^2 + 28x - 24$
- c. $8x^2 - 26x - 24$
- d. $8x^2 + 26x - 24$ ✓

$$(2x + 8)(4x - 3) = 8x^2 - 6x + 32x - 24 = 8x^2 + 26x - 24.$$

48 Expand $(3x - 3)(5x + 3)$.

- a. $15x^2 - 6x - 9$ ✓
- b. $15x^2 + 6x - 9$
- c. $15x^2 - 4x - 9$
- d. $15x^2 - 6x + 9$

$$(3x - 3)(5x + 3) = 15x^2 + 9x - 15x - 9 = 15x^2 - 6x - 9.$$

49 Expand $(3x - 9)(x - 7)$.

- a. $3x^2 - 30x - 63$
- b. $3x^2 - 28x + 63$
- c. $3x^2 - 30x + 63$ ✓
- d. $3x^2 + 30x + 63$

$$(3x - 9)(x - 7) = 3x^2 - 21x - 9x + 63 = 3x^2 - 30x + 63.$$

50 Expand $(3x + 7)(2x + 4)$.

a. $6x^2 - 26x + 28$

b. $6x^2 + 26x + 28$ ✓

c. $6x^2 + 26x - 28$

d. $6x^2 + 28x + 28$

$$(3x + 7)(2x + 4) = 6x^2 + 12x + 14x + 28 = 6x^2 + 26x + 28.$$

51 Expand $(2x + 9)(2x - 8)$.

a. $4x^2 + 2x - 72$ ✓

b. $4x^2 - 2x - 72$

c. $4x^2 + 4x - 72$

d. $4x^2 + 2x + 72$

$$(2x + 9)(2x - 8) = 4x^2 - 16x + 18x - 72 = 4x^2 + 2x - 72.$$

Multiple choice — hard 15

52 Find the remainder when $f(x) = x^3 + 5x^2 + 2x + 3$ is divided by $(x - (-2))$, using the remainder theorem.

a. 11 ✓

b. 9

c. 13

d. -11

By the remainder theorem, the remainder equals $f(-2)$:

$$f(-2) = (-2)^3 + (5)(-2)^2 + (2)(-2) + (3) = 11.$$

- 53 Find the remainder when $f(x) = x^3 - 6x^2 - 7x - 15$ is divided by $(x - (2))$, using the remainder theorem.

a. -45 ✓

b. 45

c. -43

d. -47

By the remainder theorem, the remainder equals $f(2)$: $f(2) = (2)^3 + (-6)(2)^2 + (-7)(2) + (-15) = -45$.

- 54 Find the remainder when $f(x) = x^3 + 3x^2 + 9$ is divided by $(x - (-4))$, using the remainder theorem.

a. -5

b. -7 ✓

c. -9

d. 7

By the remainder theorem, the remainder equals $f(-4)$:

$$f(-4) = (-4)^3 + (3)(-4)^2 + (0)(-4) + (9) = -7.$$

- 55 Find the remainder when $f(x) = x^3 + 3x^2 + 6x + 14$ is divided by $(x - (-1))$, using the remainder theorem.

a. 12

b. 10 ✓

c. -10

d. 8

By the remainder theorem, the remainder equals $f(-1)$:

$$f(-1) = (-1)^3 + (3)(-1)^2 + (6)(-1) + (14) = 10.$$

- 56 Find the remainder when $f(x) = x^3 - 4x^2 - 4x - 15$ is divided by $(x - (-1))$, using the remainder theorem.

a. -16 ✓

b. -14

c. -18

d. 16

By the remainder theorem, the remainder equals $f(-1)$:

$$f(-1) = (-1)^3 + (-4)(-1)^2 + (-4)(-1) + (-15) = -16.$$

- 57 An investment of 3000 dollars grows according to $A(t) = 3000(1.07)^t$, where t is the number of years. Find the value of the investment after 5 years, to the nearest dollar.

a. 4408

b. 4008

c. 4208 ✓

d. 6311

$$A(5) = 3000(1.07)^5 \approx 4208 \text{ dollars.}$$

- 58 An investment of 8000 dollars grows according to $A(t) = 8000(1.08)^t$, where t is the number of years. Find the value of the investment after 4 years, to the nearest dollar.

a. 11084

b. 10884 ✓

c. 10684

d. 16326

$$A(4) = 8000(1.08)^4 \approx 10884 \text{ dollars.}$$

- 59 An investment of 3000 dollars grows according to $A(t) = 3000(1.07)^t$, where t is the number of years. Find the value of the investment after 8 years, to the nearest dollar.

a. 5155 ✓

b. 4955

c. 7732

d. 5355

$$A(8) = 3000(1.07)^8 \approx 5155 \text{ dollars.}$$

- 60 An investment of 5000 dollars grows according to $A(t) = 5000(1.05)^t$, where t is the number of years. Find the value of the investment after 6 years, to the nearest dollar.

a. 6700 ✓

b. 6900

c. 6500

d. 10051

$$A(6) = 5000(1.05)^6 \approx 6700 \text{ dollars.}$$

- 61 An investment of 8000 dollars grows according to $A(t) = 8000(1.03)^t$, where t is the number of years. Find the value of the investment after 8 years, to the nearest dollar.

a. 10334

b. 10134 ✓

c. 15201

d. 9934

$$A(8) = 8000(1.03)^8 \approx 10134 \text{ dollars.}$$

- 62 Solve $x^3 + 3x^2 - 10x = 0$ by factoring completely. Excluding the solution $x = 0$, find the other two solutions as an ordered pair (smaller first).

- a. $(-5, 1)$
- b. $(-4, 2)$
- c. $(-5, 2)$ ✓
- d. $(5, -2)$

Factor out x : $x(x^2 + 3x - 10) = 0$, then factor the quadratic: $x(x - (-5))(x - (2)) = 0$. By the zero product property, $x = 0$, $x = -5$, or $x = 2$. Excluding 0: $(-5, 2)$.

- 63 Solve $x^3 + 11x^2 + 18x = 0$ by factoring completely. Excluding the solution $x = 0$, find the other two solutions as an ordered pair (smaller first).

- a. $(9, 2)$
- b. $(-9, -2)$ ✓
- c. $(-9, -3)$
- d. $(-8, -2)$

Factor out x : $x(x^2 + 11x + 18) = 0$, then factor the quadratic: $x(x - (-9))(x - (-2)) = 0$. By the zero product property, $x = 0$, $x = -9$, or $x = -2$. Excluding 0: $(-9, -2)$.

- 64 Solve $x^3 + 3x^2 - 40x = 0$ by factoring completely. Excluding the solution $x = 0$, find the other two solutions as an ordered pair (smaller first).

- a. $(-8, 4)$
- b. $(8, -5)$
- c. $(-8, 5)$ ✓
- d. $(-7, 5)$

Factor out x : $x(x^2 + 3x - 40) = 0$, then factor the quadratic: $x(x - (-8))(x - (5)) = 0$. By the zero product property, $x = 0$, $x = -8$, or $x = 5$. Excluding 0: $(-8, 5)$.

- 65 Solve $x^3 + 7x^2 - 18x = 0$ by factoring completely. Excluding the solution $x = 0$, find the other two solutions as an ordered pair (smaller first).

a. $(9, -2)$

b. $(-9, 1)$

c. $(-9, 2)$ ✓

d. $(-8, 2)$

Factor out x : $x(x^2 + 7x - 18) = 0$, then factor the quadratic: $x(x - (2))(x - (-9)) = 0$. By the zero product property, $x = 0$, $x = 2$, or $x = -9$. Excluding 0: $(-9, 2)$.

- 66 Solve $x^3 - 15x^2 + 56x = 0$ by factoring completely. Excluding the solution $x = 0$, find the other two solutions as an ordered pair (smaller first).

a. $(7, 8)$ ✓

b. $(7, 7)$

c. $(-7, -8)$

d. $(8, 8)$

Factor out x : $x(x^2 - 15x + 56) = 0$, then factor the quadratic: $x(x - (7))(x - (8)) = 0$. By the zero product property, $x = 0$, $x = 7$, or $x = 8$. Excluding 0: $(7, 8)$.